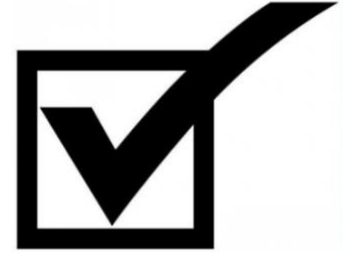




**BUSINESS
PROFESSIONALS**
of **AMERICA**
Giving Purpose to Potential



MANAGERIAL ACCOUNTING (135)

NATIONAL 2022

CONCEPT KNOWLEDGE:

Multiple Choice (15 @ 2 points each) _____ (30 points)

APPLICATION KNOWLEDGE:

Short Answer _____ (51 points)

Problem 1: Incremental Analysis _____ (25 points)

Problem 2: Capital Investment Analysis _____ (35 points)

TOTAL POINTS _____ (***141 points***)

Test Time: 90 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

EXAM GUIDELINES:

You have been hired as a Financial Assistant and will be keeping the accounting records for Digital Solutions, located at 700 Morse Road, Suite 201, Columbus, Ohio 43214. Digital Solutions provides accounting and other financial services for clients. You will complete jobs for Digital Solutions' own accounting records, as well as for clients.

You will have 90 minutes to complete your work. The test is divided into two parts: concept knowledge and application knowledge.

Your name and/or school name should *not* appear on any work you submit for grading. Write your Member ID in the provided space. Staple all pages in order before you turn in your test.

Assumptions to make when taking this assessment:

- Round all calculations to two decimal places at the final step.
- Round all percentages to one decimal place.
- Use 360 days for interest calculations.

MULTIPLE CHOICE (10 @ 2 points each = 20 points)

- | | | |
|------|-------|-------|
| 1. B | 6. A | 11. D |
| 2. C | 7. B | 12. B |
| 3. A | 8. A | 13. C |
| 4. A | 9. B | 14. B |
| 5. A | 10. A | 15. B |

Short Answer (Each answer worth 3 points; total 51 Points)

1. Stark Industries has expected sales ranging from \$170,000 to \$200,000. Variable costs and their percentage relationships are: Sales commissions 5%, Advertising 4%, Traveling 3%, and Hair Supplies 8%. Fixed expenses are Salaries \$34,000, Depreciation \$7,000, and Insurance \$1,000.

If sales are \$180,000, what are total costs?	\$78,000
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Actual expenses for Stark Industries are as follows:

Variable Expense		Fixed Expenses	
Sales Commissions	\$9,200	Salaries	\$34,000
Advertising	7,000	Depreciation	7,000
Travel	5,100	Insurance	1,000
Hair Supplies	16,800		

If sales are \$180,000 what is the budget variance for total expenses?	\$2,100
Is it Favorable or Unfavorable?	Unfavorable

2. Acme Corp. has budgeted three hours of direct labor per anvil, at a standard cost of \$17 per hour. During August, laborers actually worked 189 hours completing 84 anvils. All 84 anvils actually produced were sold. Acme paid the laborers \$18 per hour.

Calculate the following variances and identify if each was favorable (F) or unfavorable (U).

	Variance	Favorable or Unfavorable
Labor Rate Variance	\$189	Unfavorable
Labor Efficiency Variance	\$1,071	Favorable
Total Labor Spending Variance	\$882	Favorable

3. Bluth Industries manufactures kiosks for shopping malls. During the second quarter of the year, the company manufactured 3,500 kiosks, using 22,500 pounds of plastic. The plastic cost the company \$117,100. According to the standard cost, each kiosk should require 6 pounds of plastic at a cost of \$8 per pound.

Calculate the following variances and identify if each was favorable (F) or unfavorable (U).

	Variance	Favorable or Unfavorable
Materials Price Variance	\$62,900	F
Materials Quantity Variance	\$12,000	U
Total Materials Spending Variance	\$50,900	F

4. Central Perk provides the following financial information:

Income from operations	\$200,000
Interest expense	50,000
Gains/(losses) on sale of equipment	3,000
Net income	147,000
Total assets at Jan. 1	2,900,000
Total assets at Dec .31	3,300,000
Target Rate of Return	8%

Calculate return on investment	4.7%
Calculate the Residual Income	(\$101,000)

Problem 1 (Each answer worth 5 points; total 25 Points)

- A. It costs Wonka Inc. \$45 per unit (\$27 variable and \$18 fixed) to make a product that normally sells for \$80. A wholesaler is offering to buy 5,000 units at \$30 each. Wonka will incur special shipping costs of \$1 per unit.

Assuming that Wonka has excess capacity how much income (loss) would Wonka realize by accepting this special order?	\$10,000
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- B. Duff's Corp. incurs unit costs of \$13 (\$8 variable & \$5 fixed) in making a part for its product. A supplier has offered to make 10,000 of the parts at \$7.30 per unit. If the offer is accepted, Duff's will save all variable costs but no fixed costs.

How much money will Duff's save (lose) by buying the part?	\$7,000
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- C. Wayne Industries makes an unassembled product that it currently sells for \$40. Production costs are \$28. Wayne is considering assembling the product and selling it for \$68. The cost to assemble the product is \$12.

Should Wayne sell <i>before assembly</i> or <i>process further</i> ?	Process further
If Wayne decides to process further, net income will increase (decrease) by how much per unit?	\$16

- D. Spacely Space Sprockets manufactures three types of sprockets – the Cosmo, the Astro, and the Rosie. The company is deciding whether to drop the Rosie product line because it has an operating loss.

	Cosmo	Astro	Rosie
Sales revenue	\$80,000	\$40,000	\$30,000
Variable costs	(40,000)	(15,000)	(12,000)
Contribution margin	\$40,000	\$25,000	\$18,000
Fixed costs	(15,000)	(5,000)	(24,000)
Operating income (loss)	\$25,000	\$20,000	\$(6,000)

Assuming fixed costs are unavoidable, if Spacely drops the Rosie product line and does not replace it, what effect will this have on operating income?	(\$18,000)
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Problem 2 (Each answer worth 5 points; total 35 Points)

Grey-Sloan Memorial is evaluating two possible investments in depreciable plant assets. The company uses the straight-line method of depreciation. The following information is available:

	Investment A	Investment B
Initial capital investment	\$108,000	\$151,000
Estimated useful life	10 years	10 years
Estimated residual value	0	\$23,000
Estimated annual cash inflow for 10 years	\$30,000	\$45,000
Estimated annual cash outflow for 10 years	\$5,000	\$7,000
Required overhaul	\$20,000 at the end of 5 years	\$15,000 every 4 years
Required rate of return	10%	14%

For each investment, calculate the following:

	Investment A	Investment B
Payback Period	4.32 years	3.97 years
Accounting Rate of Return/Simple Rate of Return	22.6%	25.5%
Net Present Value	\$33,205	\$39,273

Which investment is preferable? **Investment B**